

1.Armstrong number:

```
#include <stdio.h>
```

```
int main()
{
    int n,count=0;
    scanf("%d",&n);
    int x=n,y=n;
    do{
        count++;
        n/=10;
    }while(n!=0);
    int d=0,ans=0,a=1;
    while(x!=0){
        d=x%10;
        for(int i=0;i<count;i++){
            a*=d;

        }ans+=a;
        a=1;
        x/=10;
    }if(y==ans){
        printf("Armstrong number");
    }else{
        printf("not an Armstrong number");
    }

    return 0;
}
```

2.Magic number:

```

#include <stdio.h>

int sum=0;

int magic(int num);

int main()
{
    int n;
    scanf("%d",&n);
    int x=n;
    int a=magic(x);
    while(a>9){
        a=magic(a);
    } printf("%d\n",a);
    if(a==7){
        printf("Magic Number");
    }else{
        printf("Not a Magic number");
    }
    return 0;
}

int magic(int num){
    int d=0;
    sum=0;
    while(num!=0){
        d=num%10;
        sum+=d;
        num/=10;
    }
    return sum;
}

```

3.perfect,abundant,deficient number:

```

#include <stdio.h>

```

```

int main()
{
    int n,sum=0;
    scanf("%d",&n);
    for(int i=1;i<=n/2;i++){
        if(n%i==0){
            sum+=i;
        }
    }
    if(sum==n){
        printf("perfect number");
    }else if(sum>n){
        printf("abundant number");
    }else{
        printf("deficient number");
    }

    return 0;
}

```

4.prime digits:

```
#include <stdio.h>
```

```

int main()
{
    int n;
    scanf("%d",&n);
    int x=n;

    int d=0,c1=0,c2=0;
    while(x!=0){
        d=x%10;
    }
}

```

```
    if(d==1 || d==3 || d==5 || d==7){  
        c1++;  
    }  
    x/=10;  
} printf("%d",c1);  
  
return 0;  
}
```